

Optimal control applied to single-seater dynamics

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LGAC: Matemáticas: Análisis, Optimización, Control y sus Aplicaciones

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Outline

- 1 Introduction & Problem Formulation
- 2 Vehicle Dynamics & Constraints
- 3 Optimization & Control Theory
- 4 Numerical Solution

The Optimal Control Problem

Defining the Boundary Limits

The primary goal in motorsport is to minimize the total lap time t_f .

An Optimal Control Problem (OCP) is defined by:

$$\min_{\mathbf{x}(\cdot), \mathbf{u}(\cdot), t_f} J = \Phi(x(t_f), t_f) + \int_{t_0}^{t_f} L(\mathbf{x}(t), \mathbf{u}(t), t) dt$$

s.t.

$$\begin{aligned} \dot{\mathbf{x}}(t) &= f(\mathbf{x}(t), \mathbf{u}(t), t), \quad \mathbf{x}(t_0) = \mathbf{x}_0 \\ \mathbf{x}(t_f) &\in \mathcal{X}_f, \quad \mathbf{u}(t) \in \mathcal{U}, \quad t_f \in [t_{min}, t_{max}] \end{aligned}$$

In this case, the problem is formulated as a continuous-time optimal control problem:

$$J = \int_0^{t_f} L(\mathbf{x}(t), \mathbf{u}(t), t) dt$$

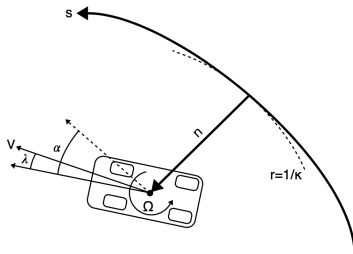
where

- L is the running cost
- **State Variables ($\mathbf{x}$):** Define the vehicle's condition at any point (e.g., position x, y , velocity v , heading ψ , yaw rate Ω).
- **Control Variables ($\mathbf{u}$):** The inputs manipulated by the driver/solver (e.g., steering angle δ , throttle command, braking pressure).
- **Constraints:** The system is strictly bounded by the track limits and the physical capabilities of the vehicle.

Which in this case, the running cost is simply 1, representing the time elapsed. The goal is to minimize the total lap time t_f . Then

$$J = \int_0^{t_f} 1 \, dt = t_f$$

- **Longitudinal Position (s):** The distance along the center line.
- η : The lateral deviation perpendicular from the center line.
- α : The angle between the vehicle's heading and the tangent to the center line.



Then we have

$$\dot{s} = \frac{V \cos(\alpha - \lambda)}{1 - \eta \kappa}$$

$$\dot{\eta} = V \sin(\alpha - \lambda)$$

$$\dot{\alpha} = \omega - \frac{\kappa V \cos(\alpha - \lambda)}{1 - \eta \kappa}$$

where κ is the curvature of the track at the current position, λ is the slip angle of the vehicle and $1 - \eta\kappa > 0$ is the jacobian of the transformation of the coordinates.

The change of variables from time to space is given by

$$\frac{d}{dt} = \dot{s} \frac{d}{ds}$$

and changing all the temporal derivatives to spatial derivatives

$$\frac{dV}{ds} = \frac{\dot{V}}{\dot{s}}, \quad \frac{d\eta}{ds} = \frac{\dot{\eta}}{\dot{s}}, \quad \frac{d\alpha}{ds} = \frac{\dot{\alpha}}{\dot{s}}, \quad \frac{d\omega}{ds} = \frac{\dot{\omega}}{\dot{s}}, \dots$$

This parametrization allows us to express the total lap time as a function of the spatial distance s :

$$t_f = \int_0^{t_f} dt = \int_0^S \frac{1}{\dot{s}} ds$$

where S is the total length of the track.

Vehicle Modeling Approaches

Capturing Chassis Behavior

Point-Mass Model

- Treats the vehicle as a single particle.
- Useful for rapid trajectory planning, but assumes instantaneous directional changes.
- Ignores vehicle orientation and rotational inertia.

Planar Models (Bicycle / 4-Wheel)

- Essential for capturing accurate F1 behavior.
- Introduces critical dynamic variables:
 - **Yaw Rate (Ω)**: The rotation of the car around its vertical axis.
 - **Slip Angles (λ)**: The difference between the direction the wheel is pointing and the direction it is actually traveling.
- The car must physically rotate before it can corner effectively.

Aerodynamic Dependency

The Speed-Grip Coupling

- F1 dynamics are dominated by aerodynamic downforce, which couples lateral grip to longitudinal speed.
- **Downforce Equation:**

$$F_z = \frac{1}{2}\rho C_L A v^2$$

- Because normal load (F_z) increases quadratically with velocity (v), the car possesses significantly more cornering potential at high speeds.
- This creates a highly non-linear optimization problem: the optimal braking and turn-in points shift continuously based on the aero-map.

Tire Dynamics and Limits

The Friction Ellipse

- Tires possess a finite amount of grip, constrained by the coefficient of friction (μ) and the vertical load (F_z).
- The combined forces must remain within the Friction Ellipse boundary:

$$F_{total} = \sqrt{F_{long}^2 + F_{lat}^2} \leq \mu F_z$$

- **The Trade-off:** Accelerating or braking (F_{long}) inherently reduces the available lateral grip (F_{lat}) required for cornering.
- The solver must constantly balance the tire's capacity at the absolute limit.

Indirect Methods and Theory

Pontryagin's Minimum Principle

The **indirect methods** rely on the Calculus of Variations. And the central question is: What is the trajectory $\mathbf{x}^*(t)$, $\mathbf{u}^*(t)$ that minimizes the cost functional J while satisfying the system dynamics and constraints?

We define the Hamiltonian function incorporating the system dynamics and the costates λ :

$$H(\mathbf{x}, \mathbf{u}, \lambda, t) = L(\mathbf{x}, \mathbf{u}, t) + \lambda^T f(\mathbf{x}, \mathbf{u}, t)$$

Then the augmented cost functional becomes

$$J_a = \Phi(\mathbf{x}(t_f), t_f) + \int_{t_0}^{t_f} \left[L(\mathbf{x}, \mathbf{u}, t) + \lambda^T (f(\mathbf{x}, \mathbf{u}, t) - \dot{\mathbf{x}}) \right] dt$$

The necessary conditions for optimality are derived from the Hamiltonian:

$$\begin{aligned}\dot{\mathbf{x}} &= \frac{\partial H}{\partial \boldsymbol{\lambda}} = f(\mathbf{x}, \mathbf{u}, t) \\ \dot{\boldsymbol{\lambda}} &= -\frac{\partial H}{\partial \mathbf{x}} \\ 0 &= \frac{\partial H}{\partial \mathbf{u}}\end{aligned}$$

If the control $\mathbf{u}$ is bounded, the stationarity condition transforms into the **Pontryagin's Minimum Principle**:

$$\mathbf{u}^*(t) = \arg \min_{\mathbf{u} \in \mathcal{U}} H(\mathbf{x}^*(t), \mathbf{u}, \boldsymbol{\lambda}^*(t), t)$$

For this specific problem, $\mathbf{x} = [\eta, \alpha, V, \omega, a_x, a_y, t]^T$,
 $\mathbf{u} = [\delta, Thrust]^T$, variable independent s , and the running cost
 $L = 1/\dot{s}$, where

- η is the lateral deviation from the track centerline
- α is the heading error
- V is the vehicle speed
- ω is the yaw rate
- a_x is the longitudinal acceleration
- a_y is the lateral acceleration
- δ is the steering angle
- $Thrust$ is the longitudinal force applied by the powertrain

The Hamiltonian is defined as

$$H = \frac{1}{\dot{s}} + \lambda_{\eta} \dot{\eta} + \lambda_{\alpha} \dot{\alpha} + \lambda_V \dot{V} + \lambda_{\omega} \dot{\omega} + \lambda_{a_x} \dot{a}_x + \lambda_{a_y} \dot{a}_y + \lambda_t \dot{t}$$

where $\boldsymbol{\lambda} = [\lambda_{\eta}, \lambda_{\alpha}, \lambda_V, \lambda_{\omega}, \lambda_{a_x}, \lambda_{a_y}, \lambda_t]^T$ are the costates associated with each state variable.

- This creates a Two-Point Boundary Value Problem (TPBVP).
- The solver is extremely sensitive to the initial guess of the costates (λ), which have no physical intuition, making it highly unstable for complex vehicle dynamics.

Direct Method

From the OCP to a Numerical Optimization Problem

Can we avoid the sensitivity to initial guesses and the complexity of solving a TPBVP?

Yes. **Direct methods** discretize the continuous optimal control problem and transform it into a finite-dimensional **Nonlinear Programming (NLP)** problem.

The OCP searches for optimal trajectories

$$\mathbf{x}^*(\cdot), \quad \mathbf{u}^*(\cdot)$$

which are functions defined over the spatial domain $s \in [0, S]$.

That's why

$$\mathbf{x} : [0, S] \rightarrow \mathbb{R}^{n_x}, \quad \mathbf{u} : [0, S] \rightarrow \mathbb{R}^{n_u}.$$

Therefore, the optimization is infinite-dimensional.

The direct method replaces these unknown functions by finite-dimensional approximations.

Finite-Dimensional Approximation

We approximate the continuous trajectories by finite-dimensional representations:

$$\mathbf{x}(s) \approx \mathbf{x}_N(s), \quad \mathbf{u}(s) \approx \mathbf{u}_N(s).$$

In general, we can write

$$\mathbf{x}_N(s) = \sum_{i=0}^N \mathbf{x}_i \phi_i(s)$$

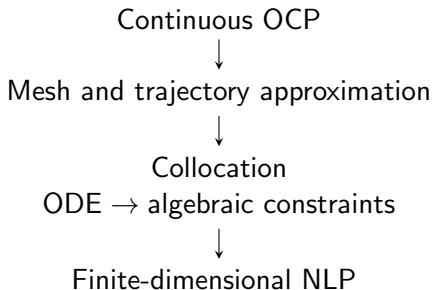
$$\mathbf{u}_N(s) = \sum_{i=0}^N \mathbf{u}_i \psi_i(s)$$

Finite-Dimensional Approximation

- $\mathbf{x}_N(s), \mathbf{u}_N(s)$: approximations of the continuous trajectories.
- $\mathbf{x}_i, \mathbf{u}_i$: finite-dimensional coefficients or nodal values.
- $\phi_i(s), \psi_i(s)$: known basis functions.
- The unknown functions are therefore replaced by a finite set of unknown numbers.

Transcription

The general idea of the transcription is to convert the continuous-time/spatial OCP into a finite-dimensional NLP.



- Discretization: Choose a mesh

$$0 = s_0 < s_1 < \cdots < s_N = S$$

- Approximation: Represent the continuous trajectories using a finite number of variables

$$\mathbf{x}(s) \approx \mathbf{x}_N(s), \quad \mathbf{u}(s) \approx \mathbf{u}_N(s)$$

- Collocation: Enforce the system dynamics at selected points, transforming differential equations into algebraic constraints.

Our Transcription

Gauss-Lobatto and Hermite-Simpson

There are a lot of transcription methods in this case we use a third-order Gauss-Lobatto transcription, the collocation points on each interval are

$$\tau_0 = -1, \quad \tau_1 = 0, \quad \tau_2 = 1.$$

This leads to the classical **Hermite-Simpson collocation scheme**.

Gauss-Lobatto (order 3) $\implies$ Hermite-Simpson
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Hermite-Simpson

Local Representation

Consider a single mesh interval

$$[s_i, s_{i+1}], \quad h_i = s_{i+1} - s_i.$$

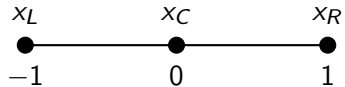
Introduce the normalized coordinate

$$s(\tau) = \frac{s_i + s_{i+1}}{2} + \frac{h_i}{2}\tau, \quad \tau \in [-1, 1].$$

Therefore,

$$\tau = -1 \Rightarrow s = s_i, \quad \tau = 0 \Rightarrow s = s_c, \quad \tau = 1 \Rightarrow s = s_{i+1}.$$

Visually, it can be represented as follows:



Hermite-Simpson

Cubic State Approximation

On each interval, the state trajectory is approximated by a cubic Hermite polynomial.

$$x_H(\tau) = H_{00}(\tau)x_L + H_{10}(\tau)\frac{h}{2}f_L + H_{01}(\tau)x_R + H_{11}(\tau)\frac{h}{2}f_R$$

where

$$f_L = f(x_L, u_L, s_L) = x'_L, \quad f_R = f(x_R, u_R, s_R) = x'_R.$$

Hermite Basis Functions

The cubic Hermite basis functions on $\tau \in [-1, 1]$ are

$$H_{00}(\tau) = \frac{1}{4}(1 - \tau)^2(2 + \tau),$$

$$H_{10}(\tau) = \frac{1}{4}(1 - \tau)^2(1 + \tau),$$

$$H_{01}(\tau) = \frac{1}{4}(1 + \tau)^2(2 - \tau),$$

$$H_{11}(\tau) = -\frac{1}{4}(1 + \tau)^2(1 - \tau).$$

Hermite Basis Functions

Their role is

Basis function	Associated quantity
H_{00}	x_L
H_{10}	$\frac{h}{2}x'_L$
H_{01}	x_R
H_{11}	$\frac{h}{2}x'_R$

Hence,

$$x_H(\tau) = \sum_j H_j(\tau) c_j.$$

From the ODE to a Collocation Constraint

The continuous dynamics are

$$\frac{dx}{ds} = f(x, u, s).$$

Integrating over one mesh interval gives

$$x_R - x_L = \int_{s_L}^{s_R} f(x(s), u(s), s) ds.$$

The integral is approximated using Simpson's rule:

$$\int_{s_L}^{s_R} f(s) ds \approx \frac{h}{6} (f_L + 4f_C + f_R).$$

Hermite-Simpson

Collocation Constraint

Combining the integrated dynamics with Simpson's rule:

$$x_R - x_L = \frac{h}{6} (f_L + 4f_C + f_R).$$

Therefore,

$$g_{\text{dyn}} = x_R - x_L - \frac{h}{6} (f_L + 4f_C + f_R) = 0.$$

The differential equation has been transformed into an **algebraic equality constraint**.

Collocation Constraint

Meaning of the Terms

$$x_L = x(s_L),$$

$$x_R = x(s_R),$$

$$f_L = f(x_L, u_L, s_L),$$

$$f_R = f(x_R, u_R, s_R),$$

$$f_C = f(x_C, u_C, s_C),$$

$$h = s_R - s_L.$$

Therefore,

$$x_R - x_L - \frac{h}{6}(f_L + 4f_C + f_R) = 0$$

constrains the neighboring state values to be consistent with the vehicle dynamics.

From the Continuous OCP to Algebraic Constraints

Continuous dynamics

$$x'(s) = f(x, u, s)$$



Integrate over $[s_L, s_R]$

$$x_R - x_L = \int_{s_L}^{s_R} f \, ds$$



Simpson quadrature

$$\int f \, ds \approx \frac{h}{6} (f_L + 4f_C + f_R)$$



Collocation constraint

$$x_R - x_L - \frac{h}{6} (f_L + 4f_C + f_R) = 0$$

Hermite-Simpson

Putting the Pieces Together

Hermite interpolation provides a smooth approximation of the state trajectory inside each interval:

$$x(s) \approx x_H(s).$$

Simpson quadrature approximates the integral of the dynamics:

$$\int_{s_L}^{s_R} f(x, u, s) ds \approx \frac{h}{6} (f_L + 4f_C + f_R).$$

Therefore,

Hermite interpolation + Simpson quadrature = Hermite-Simpson collocation

From OCP to NLP

After transcription, the decision variables are finite-dimensional:

$$z = \{x_i, u_i\}_{i=0}^N.$$

The continuous optimal control problem becomes

$$\begin{array}{ll} \min_z & J_N(z) \\ \text{s.t.} & g(z) = 0, \\ & h(z) \leq 0. \end{array}$$

Our Optimal Lap-Time NLP

For our vehicle model,

$$\mathbf{x} = [\eta, \alpha, V, \omega, a_x, a_y, t]^T$$

and

$$\mathbf{u} = [\delta, \text{Thrust}]^T.$$

The decision vector contains the discretized state and control trajectories:

$$z = \{\mathbf{x}_i, \mathbf{u}_i\}_{i=0}^N.$$

The objective is the lap time:

$$\min_z t(S)$$

Our NLP

Constraints

The NLP incorporates

- **Vehicle dynamics**

$$g_{\text{dyn}}(z) = 0.$$

- **Track boundaries**

$$\eta_{\min} \leq \eta(s) \leq \eta_{\max}.$$

- **Tire force limits**

$$\sqrt{F_{\text{long}}^2 + F_{\text{lat}}^2} \leq \mu F_z.$$

■ Powertrain constraints

$$P \leq P_{\max}.$$

■ Periodic boundary conditions

$$x(0) = x(S)$$

for the periodic states.

Numerical Solution

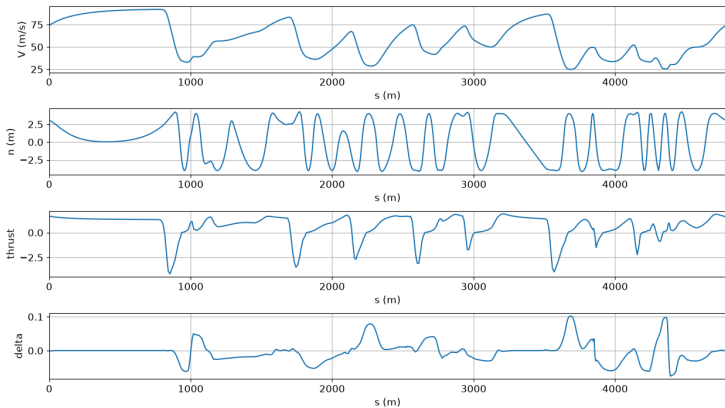
The resulting NLP is solved using the **IPOPT** solver, which implements an interior-point method for large-scale nonlinear optimization. The solver iteratively updates the decision variables z to minimize the objective function while satisfying all constraints.

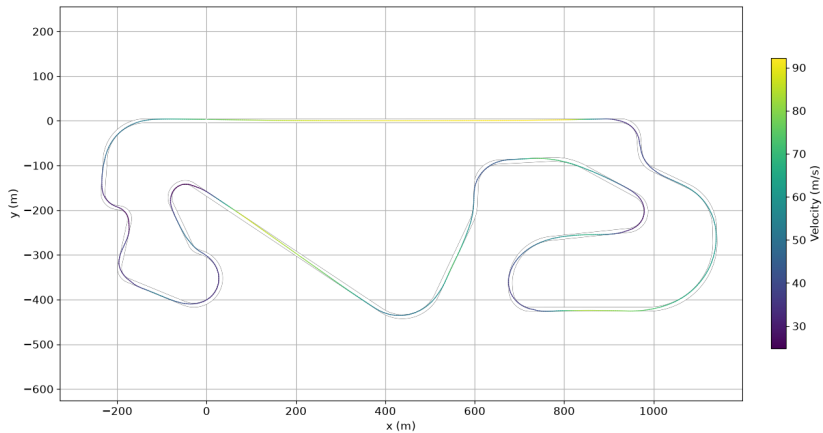
```
Total seconds in IPOPT = 89.082
```

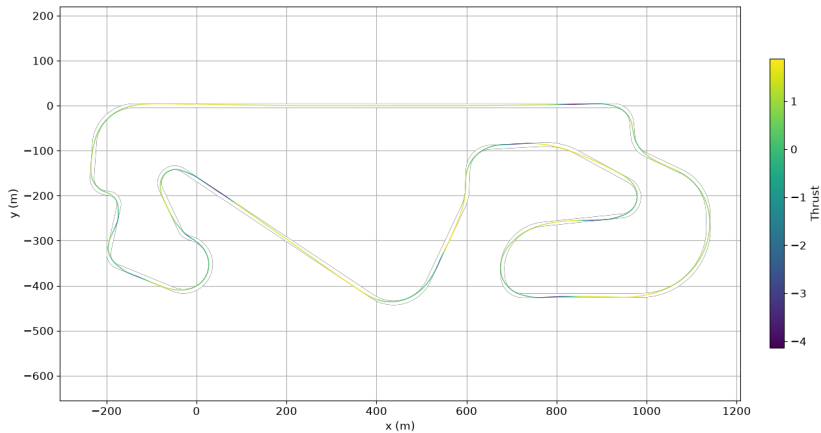
```
EXIT: Optimal Solution Found.
```

```
Optimization finished
```

```
Laptime solution: 90.259603 s
```







¡Gracias por su atención!

Este trabajo forma parte del proyecto de Ciencia Básica y Frontera SECITHI-2025 titulado:

Teoría y aplicaciones de control óptimo estocástico

Clave: **CBF-2025-I-4436**, etapa I.

Referencias I



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